

Safety Vision

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Drafted by



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Executive summary

The cargoloop operating environment will be susceptible to specific safety related hazards as well as security threats. These aspects need to be identified and understood through an integral system approach taking into account the cargoloop product development phases as well as operations. The first step towards hyperloop system safety concept is determination of the safety vision. Safety vision will provide the baseline for the development of the cargoloop safety product development plan and safety case based on the cargoloop system goal and its intended use.

This document describes the safety vision for hyperloop. The objective of the document is to present a generic approach applicable to both cargoloop and hyperloop for passengers. The combination of ensuring safe operations and continuous availability is at the core of safety design for hyperloop. To stipulate, the priority is to:

- ensure safety to human life,
- ensure safety to transported goods,
- ensure safety to environment,
- ensure safety to public/private property and
- ensure safety to hyperloop infrastructure

Secondary to this is finding a way to achieve this safety whilst also guaranteeing high availability of the system and commercial viability.

The **safety goal** for hyperloop safety system is to ensure:

- ***Safe travelling by hyperloop***
- ***Safe living near hyperloop***
- ***Safe working on hyperloop***
- ***Safe transport by cargoloop***

The **safety vision** that will provide guidance in the entire hyperloop safety design and development is:

Hyperloop is a safe-by-design system that ensures in an integrated system-manner that hazards to human life, environment, public/private property and hyperloop infrastructure are prevented, eliminated and controlled.

Safety is one of the key system properties for the final acceptance of the hyperloop not only by the regulators but most importantly by the general public. Following the safe-by-design vision, safety must be guaranteed in the design. A systematic integral approach is taken with regards to safety based on a functional analysis of the system and the risk associated with identified functions failing. This methodology is depicted in this document as well.

The document describes the background of the safe-by-design principles initially, it provides the vision as well as a methodology based on systematic integral approach. Seen that the hyperloop is still in a development phase and the fact that no regulatory framework exists this document should not be seen as a complete work but rather as a description for the first steps towards a safe hyperloop system.

Definitions

failure

termination of the ability of an item to perform the function for which it was designed

[SOURCE: ISO 14620-2:2011, definition 3.5]

ISO 16159:2012(en), 2.4

hazard

source of danger, which could harm property or personnel

risk

- (1) probability of loss or injury from a hazard [ISO 22538-4]
- (2) effect of uncertainty on objectives

risk analysis

process of examining identified risk factors for probability of occurrence, potential loss, and potential risk-handling strategies [ISO/IEC/IEEE 24765:2017]

risk identification

organized, systematic approach to determining the risk factors associated with a planned activity, project, or program [ISO/IEC/IEEE 24765:2017]

safety

freedom from unacceptable [\(3.13\)](#)

system safety

- (1) ability of a system to be free from unacceptable [\(3.13\)](#)

Note 1 to entry: A system is defined as a set or group of interacting, interrelated or interdependent elements or parts, that are organized and integrated to form a collective unity or a unified whole, to achieve a common objective. In a broader view the definition of a system consists in the hardware, [\(3.19\)](#), human systems integration, procedures and training. Therefore, system safety is part of the systems

engineering process and should systematically address all of these domains and areas in engineering and operations in a concerted fashion to prevent, eliminate and control hazards.

Note 2 to entry: A system safety concept helps the system designer(s) to model, analyze, gain awareness about, understand and eliminate the hazards, and apply controls to achieve an acceptable level of safety (3.14). The systems-based approach to safety requires the application of scientific, technical and managerial skills to hazard identification, hazard analysis, and elimination, control, or management of hazards throughout the life-cycle of a system, program, project or an activity or a product. Hazop is one of several techniques available for the identification of hazards.

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1. Introduction

The cargoloop operating environment will be susceptible to specific safety related hazards as well as security threats. These aspects need to be identified and understood through an integral system approach taking into account the cargoloop product development phases as well as operations. The first step towards hyperloop system safety concept is determination of the safety vision. Safety vision will provide the baseline for the development of the cargoloop safety product development plan and safety case based on the cargoloop system goal and its intended use.

2. Background

Hyperloop is a new means of transport with the ambition to offer safe, high-speed, zero-emissions transport for passengers and cargo. Hyperloop is an innovative integration of several existing technologies. These technologies have existed and have been implemented in practice over decades such as for example:

- Pipeline construction, utilized in oil and gas construction;
- Pylon construction and building known from infrastructure projects;
- Magnetic Levitation, originally developed by Siemens and implemented in China, South Korea and currently in construction in Japan;
- Low pressure environment provided by vacuum technology;
- Cabin pressure utilized in aviation and space engineering;

Due to the relation to the existing technologies, especially transport modes, there is a tendency to compare the hyperloop safety to the current transport technologies such as rail or aviation. For example, hyperloop should be as safe as commercial air travel ¹. Binding safety objectives to operational hours will not assure safety, especially seen the development stage of the hyperloop. Demonstrating that hyperloop is, for example, as safe as air travel would require a certain number of operational hours of a hyperloop system. Evidence-based methodology used for demonstrating, for example, the safety of self-driving cars in a closed course testing facility is not seen as suitable either due to the nature and scale of the hyperloop technology. Safe by design is the only way to assure safety.

Designing for hyperloop safety while considering the state of maturity as well as the nature of the technology requires a flexible yet systematic approach through the entire product life cycle. Following the aviation, for example, the development process could be divided into: "initial worthiness" (design for safety), "continuous worthiness" (maintenance for safety) and "Safety Management System" (quality and safety culture within an organization). The flexible approach is necessary in dealing with uncertainties in anticipating all risks already during the early stages of the design. Some of the risks may appear or become evident only at the later stages of the product development, while

¹ [Safety for the European Hyperloop Network](#)

a few at the deployment of the technology². In addition, this approach should cater for design flexibility and allow for differentiation between passenger and cargo transport:

Safety risk context should be designed specifically for hyperloop. The generic approaches used in different transport sectors can be used to analyze the risks, hazards as well as design corresponding safety levels.

3. Safety goal

There is an increasing support in various technological fields as well as transport sectors for utilizing the generic approaches in designing for safety. Functional safety design³ related to electrotechnical systems is the foundation of a large number safety engineering in various innovative sectors. These approaches rely on a well-known and established guideline that allow for adjustments and further adoption in a new technology. One such good example is the automotive safety for connected vehicles where the functional safety and its safety integrity levels for road vehicles have been defined as a derivative of the generic functional safety / Safety Integrity Level (SIL) defined in IEC 61508.

Similarly, safe by design approach is considered as suitable especially in innovative emerging new technologies⁴ where the regulatory framework is not yet in place.

The expectation is that the regulatory framework for hyperloop shall set goals to accommodate safety and interoperability of hyperloop system. These goals should address cargo and passenger hyperloop. Safety goals should be tailored to hyperloop, typically as the opposite of a hazardous event, preferably not using any quantitative measurements from the other transport systems. Once the safety goals are defined the generic approaches used in other transport modes in identifying the risks, hazards and well defined safety levels can be utilized for a safe by design hyperloop system.

Hyperloop system safety goal⁵ is to ensure:

- *Safe travelling by hyperloop*
- *Safe living near hyperloop*
- *Safe working on hyperloop*
- *Safe transport by hyperloop*

² Van de Poel I (2016) An ethical framework for evaluating experimental technology. Sci Eng Ethics 22(3):667–686. doi:10.1007/s11948-015-9724-3

³ IEC 61508

⁴ Nanotechnology

⁵ Safety ILT 2019

The preferred design strategy for system safety is to proactively eliminate, prevent the hazards rather than mitigating their effects. For example a hazard such as a loss of vehicle resulting in a loss of life or damaged goods with substantial damage to the infrastructure should be prevented. Therefore solutions eliminating, preventing a standstill of a vehicle altogether are preferred compared to solutions that would have to withstand collisions due to a vehicle standstill.

Utilizing quantitative measures from another transport mode to demonstrate the safety of the hyperloop is not possible at this phase. However, designing for certain safety levels to proactively prevent hazards can be simulated, tested and approved as acceptable. Safe by design is therefore the only way to assure safety. This will later enable an operational experience that will confirm the hyperloop system as safer than all the other transport modes.

3. Safety vision

Hyperloop safety vision encompasses both cargoloop and passenger system. The safety vision of the hyperloop is that a safe-by-design hyperloop system will be developed following the principles of system safety. System safety considers engineering as well as management principles, criteria and techniques to consider all safety aspects throughout all phases in the systems life cycle. In addition, all the constraints in terms of cost-effectiveness, operations and timeline need to be taken into account.

Hyperloop is a safe-by-design system that ensures in an integrated system-manner that hazards to human life, environment, public/private property and hyperloop infrastructure are prevented, eliminated and controlled.

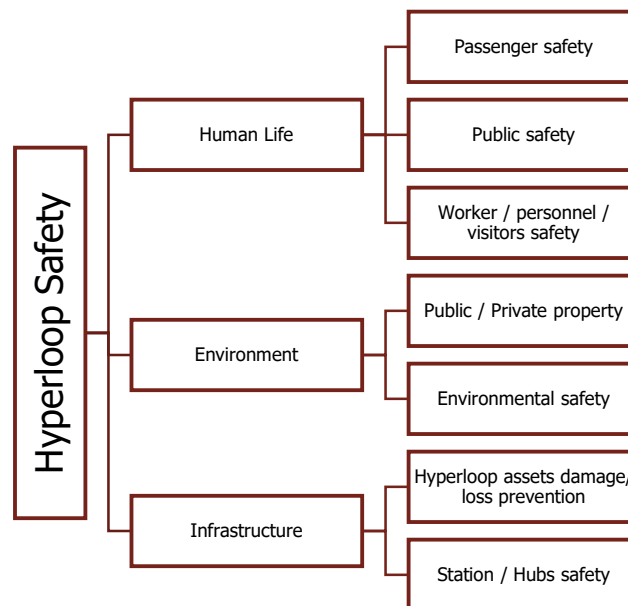


Figure 1. Illustrates key elements subject to potential risk

4. Safe by Design Hyperloop

The objective of safe by design for hyperloop is to ensure that the system safety goal is met, thus ensure safe and continuous operation for humans and environment. The combination of ensuring safe operations and continuous availability is at the core of safety design for hyperloop. To stipulate:

- the priority is to ensure safety to human life, transported goods, environment, public/private property and hyperloop infrastructure (in that specific order).
- secondary to this is finding a way to achieve this safety whilst also guaranteeing high availability of the system and commercial viability.

Several key points need to be considered and are important in achieving safe by design for hyperloop:

- The hyperloop is a new (combination of) technology, leading to high necessity of verification & validation of systems. Especially if design decisions are made which rely on redundancy, a design fault would negate the risk mitigation due to redundancy.
- The hyperloop revolves around high availability, reliability and throughput. Risks are to be managed in such a way that safety is guaranteed whilst solutions do not compromise these core characteristics.
- The hyperloop is new to the public and to ensure social acceptance a bulletproof safety design and operational demonstration is required.
- Infrastructure is a single critical link whilst vehicles are numerous and operate highly frequent. Infrastructure therefore requires a different approach than vehicle safety. However the safety design is integral and takes into account all aspects.

Based on these key-points an approach is envisioned in which systems are designed in such a way that safety is guaranteed through inherently safe design solutions which enable safe and continuous operations. To translate this to risk management strategies, this would favor strategies such as likelihood reduction and redundancy. An important note is to consider the disadvantages of chosen risk mitigation strategies (e.g. purely redundancy leaves a high vulnerability to design errors leading to a high need of verification and validation of designs).

5. Safety Design Methodology

Safety is one of the key system properties for the final acceptance of the hyperloop not only by the regulators but most importantly by the general public. Following the safe-by-design vision, safety must be guaranteed in the design. A systematic integral approach is taken with regards to safety based on a functional analysis of the system and the risk associated with identified functions failing. Safety and avoidance of failures are achieved most effectively when reliability, availability, maintainability and safety factors are continuously controlled throughout product development from its initiation through to operation instead of adding on corrective systems in later stages.

The safe-by design process for hyperloop follows the product development phases within its integral iterative approach. To structure the safe-by-design approach and integrate it into the product life-cycle a safety assessment

process is defined. This process, based on common well-known practices [ISO31000, EN50126, SAE ARP 4761, IEC61508] follows certain steps following the systems life-cycle. The phases of this process are depicted in Figure 2.

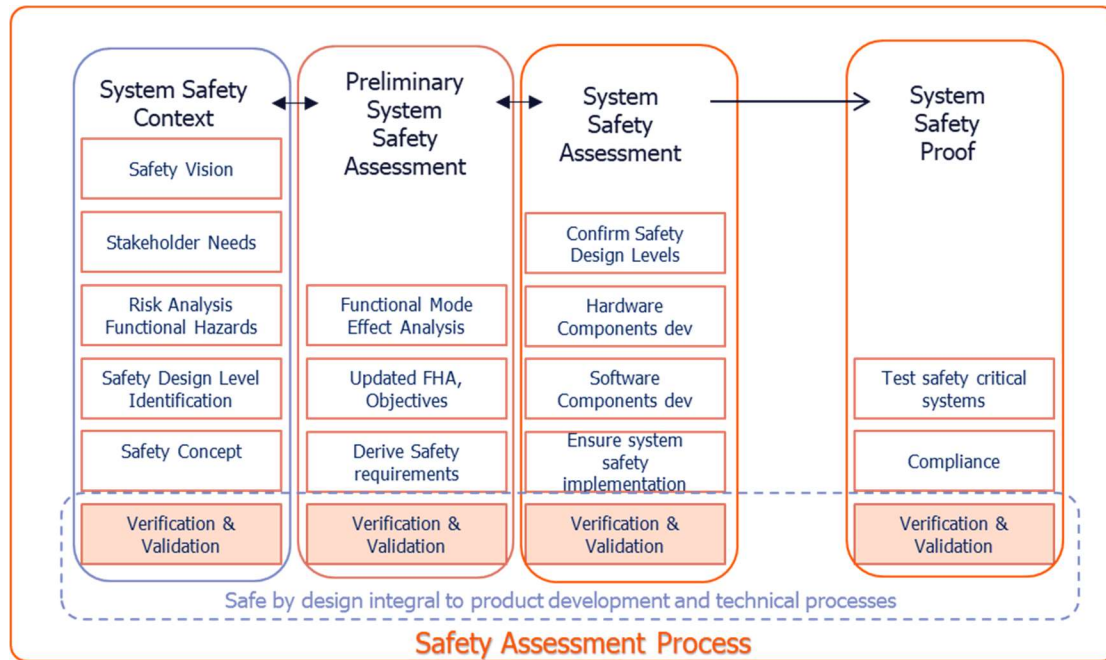


Figure 2 Hyperloop Safety Assessment Process

For each phase the deliverables will be determined based on this safety assessment process as well as the technical management process. The integration is guarded following the systems engineering approach [ISO-IEC-IEEE-15288].

The integration of the safety assessment process in the product development phases is depicted in Figure 3.

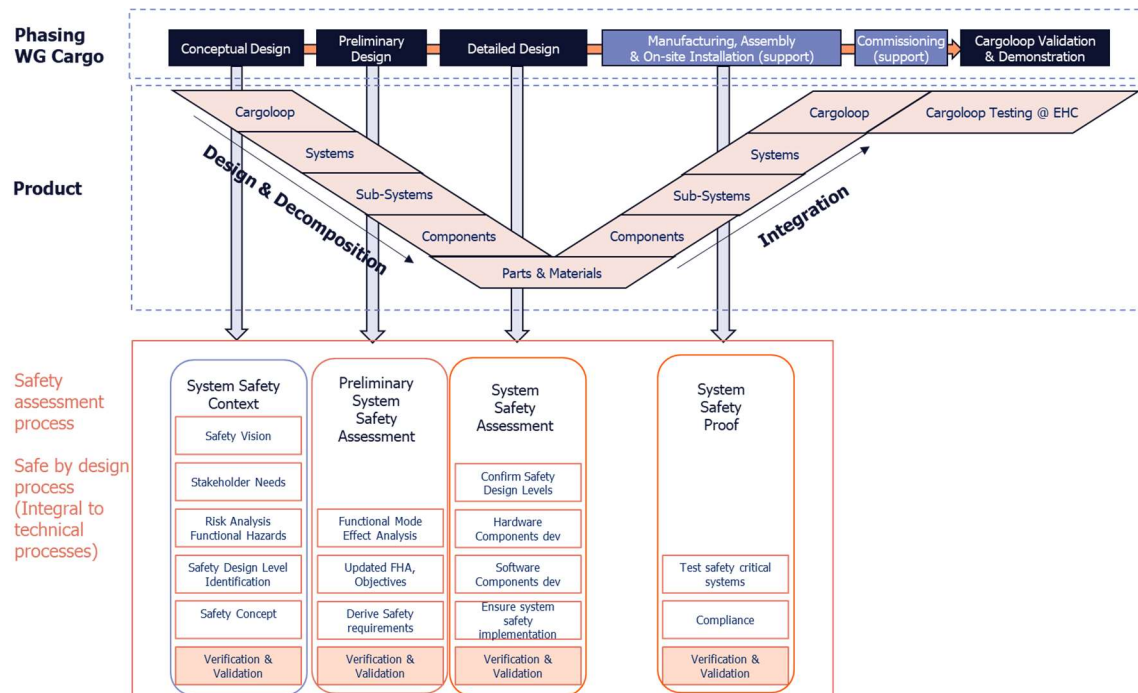


Figure 3. Safety assessment and product development

6. Systems Safety Examples

The safe by design principles have been used successfully in a number of sectors such as aviation and space engineering. This approach is utilized to prevent the accidents through hazards identification, control and elimination. Safe by design is best explained with the following sentence "Safe by design identifies the risky situations and overcome circumstances where (failure in) structure, (mal)function or (mis)use cause harm to human, environment or property." ⁶. Contrary to this a system that is designed without taking account potential current and / or future risks or failures will not be inherently safe.

A simple example is that of the Millennium Dome in London, which was designed for new year's celebration. It consists of 12 yellow towers, each 100m high. Due to their height the towers needed aircraft warning lights. In order to mitigate the risk of failure of these lights⁷, the design decision was made to install two lights on each tower.

An example of the safe by design in hyperloop is that of an automated door in a cargoloo vehicle. The door should not close while there are still obstructions: transported goods or people still in a doorway. The door should therefore be designed such that any obstruction in its closing path is detected and the closure of the door is reversed.

⁶ Incorporation of Safety into Design by Safety Cube, A Systems Engineering Perspective, M. Rajabalinejad, ICSSE 2018

⁷ <https://slideplayer.com/slide/9426217/>

Below the safety assessment in three transport sectors is described: aviation, railway and connected automated vehicles. Each relying on safe-by-design systems engineering but with different approaches. Especially the certification method is very different (see also **Error! Reference source not found.**).

6.1 Aviation

Figure 4 depicts the complex aircraft system development (SAE ARP 4754) phases and safety assessment processes (SAE ARP 4761) leading to certification. As depicted there are three basic processes relying on safe-by-design functional approach to safety:

1. Functional Hazard Assessments (FHA)
2. Preliminary System Safety Assessment (PSSA)
3. System Safety Assessment (SSA)

In addition there are also other supporting analysis such as FTA, zonal safety analysis, etc to further support safety assessment. Further, the hazards are classified into Catastrophic, Hazardous, Major and Minor failure conditions each with Design Assurance Levels (DAL). From the software perspective there are four Design Assurance Level defined (A,B,C,D) (DO-178B/C) in order to show compliance.

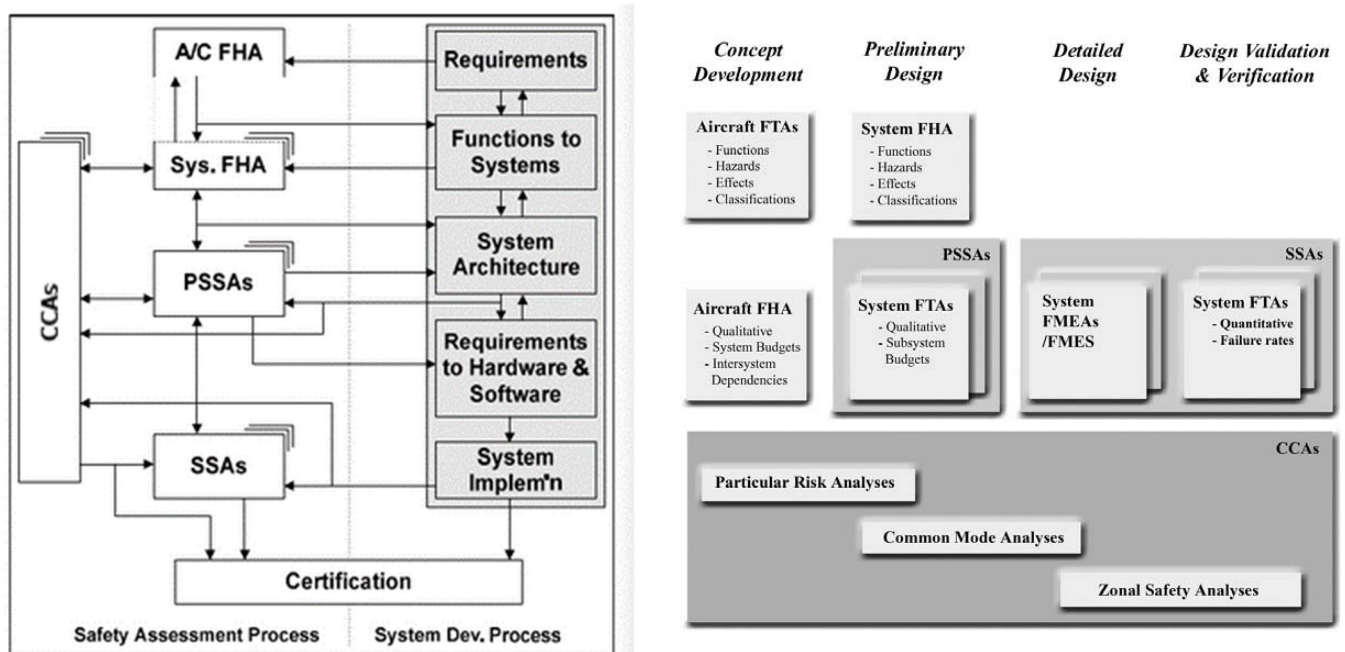


Figure 4. Aviation engineering safety assessment process for airborne systems (SAE ARP 4761)

6.2 Railway

In the railway sector it is the Common Safety Methods (CSMs) in accordance with the Article 6 of directive (EU) 2016/798 that describe how the safety levels, the achievement of safety targets and compliance with other safety requirements should be fulfilled.

The risk assessment process takes typically into account:

1. System definition
2. Identification of the hazards and related safety measures
3. Risk analysis
4. Risk evaluation
5. Definition of the system safety requirements based on the identified safety measures
6. Demonstration of system compliance with safety requirements

The results are fed into the hazard management and independent assessment / certification is performed by the independent assessor (Notified Body).

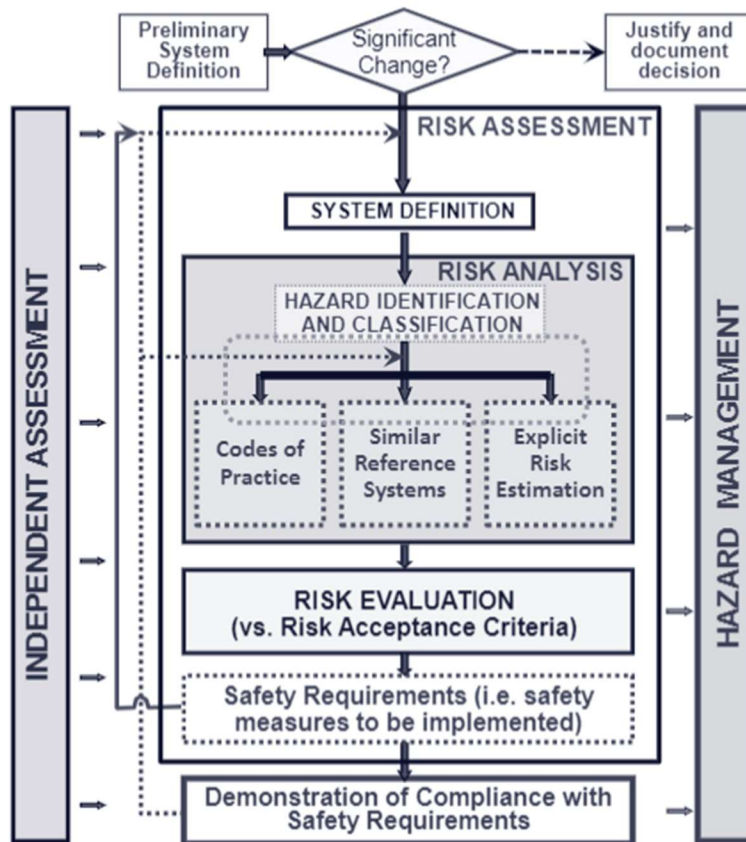
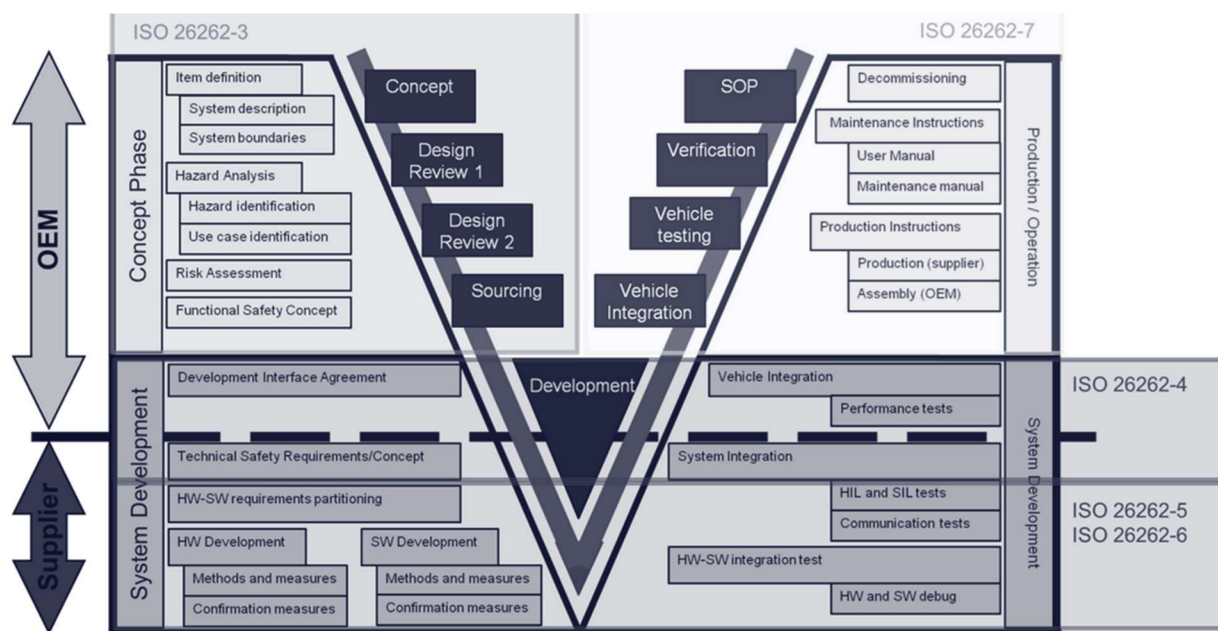


Figure 5. Railway Common Safety Methods

6.3 Connected Automated Vehicles

In industrial automation standard IEC 61508 describes the functional safety and is seen as the most widely applicable safety standard. It specifies the safety integrity level as a risk and / or failure measure. This level is based on two categories: the hardware safety integrity and systematic safety integrity. A device / system must meet the requirements coming from both these categories in order to achieve a given SIL. There are four SIL category, with one being the lowest and 4 the highest risk reduction factor.

Following this approach the automotive sector has defined automotive SIL. These levels are defined in ISO 26262 series see figure below. There are four levels ASIL A, ASIL B, ASIL C and ASIL D. These levels are established by performing a risk analysis of a potential hazard in a specific vehicle operating scenario. In addition automotive industry provided a framework "Safety First for Automated Driving" ⁸ which again shows the importance of the safe by design systems



ISO 26262
V-Modell acc. ISO-26262 with responsibilities and core tasks



Figure 6. Connect Automated Vehicles Safety, ISO26262 Applicability to system design

⁸ <https://www.daimler.com/innovation/case/autonomous/safety-first-for-automated-driving-2.html>

7. Stakeholders for safe hyperloop system

A large set of stakeholders that are vital for the hyperloop development need to be involved in the development of the safe hyperloop system in various phases of the product development. Below is a non-exhaustive list of stakeholders from various fields of expertise that will need to be involved in system safety:

- Technical management
 - Engineering as well as management department need to ensure that safety levels are set and realized in practice.
- Service providers
 - large set of existing technical processes, methods and tools such as failure description and definitions, risks, anomalies reduction, that are proven and being used by different technologies in a specific specialized domain or in several domains could be used in hyperloop safe system design as well;
 - communication infrastructure and services – for example applicability of vehicle to vehicle communications for safety design;
 - energy infrastructure;
- Suppliers of various parts of the systems
 - utilization of the specifications that already have wide acceptance within their sectors and tailor those as needed to hyperloop safety systems;
- Operators
 - Inclusion of the operational know-how and future perspectives for utilization and integration of transport modes are important in the design of hyperloop safety;
- EU/national legislative authorities
 - European Commission (EC) is working towards a EU regulatory framework for hyperloop systems which the main goal is ensuring safety and interoperability;
 - public authorities such as Ministries of Transportation, Economic Development and Innovation that unlock public funding, develop standards-receptive legislation for supporting the safety case, issue standardization mandates (through EU Committee on Standards [Regulation \(EU\) 1025/2012](#)) and public procurement, help induce public acceptance of the hyperloop systems and communicate with the EU bodies;
 - mandates the European Standardisation Organisations (ESOs) for development of the hyperloop harmonized standards in a standardisation request (mandate);
- Independent research bodies (e.g. research institutes, universities)
 - Define and validate novel safety systems unique to the hyperloop;
- Certification / inspection bodies
 - Independent assessors to validate safety testing results and certify interfaces, parts and processes for commercial use;
- Users
 - Hyperloop users/passengers' associations and logistic operators have own system safety and security requirements.



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